

Q. What are various problems which we see in society. pick any 5 problems, find out they can be solved using DSA.

Q. Trace quick sort as shown in class for the following array

1.

85	76	99	13	45	103	-5	26	31
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2.

350	790	400	200	100	96	900	890	760
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Q. Society problem

1. Traffic Navigation

DSA used

graphs,

Dijkstra's Algorithm

How it helps

Find the short & fast route for vehicles

2. Online Shopping Search

Binary Search, Hashing

Quickly finds in products & order, reducing retrieval & user waiting time information.

3. Social media friend suggestion

graphs

Represents user as nodes & suggest mutual friend

4. Library Management

Trees, Hash,

Stores & search book efficiently using ISBN.

a)

85	76	99	13	45	103	-5	26	31
i	j	i						j

$$a[i] = 85, 76, 99, \dots$$

$$a[j] = 31$$

$$a[j] > \text{pivot}$$

$$31 > 85 \times$$

$$a[i] \leq \text{pivot}$$

$$85 \leq 85$$

$$76 \leq 85$$

$$99 \leq 85 \times$$

85	76	31	99	13	45	103	-5	26	99
i	j	i		j	j	i		j	j

$$a[i] \leq \text{pivot}$$

$$85 \leq 85$$

$$76 \leq 85$$

$$31 \leq 85$$

$$13 \leq 85$$

$$45 \leq 85$$

$$103 \leq 85 \times$$

$$a[j] > \text{pivot}$$

$$99 > 85$$

$$26 > 85 \times$$

85	76	31	13	45	26	-5	103	99
i	j	i	j	j	j	j	i	j

$$a[i] \leq \text{pivot}$$

$$85 \leq 85$$

$$76 \leq 85$$

$$31 \leq 85$$

$$13 \leq 85$$

$$45 \leq 85$$

$$26 \leq 85$$

$$-5 \leq 85$$

$$99 > 85$$

$$103 > 85$$

$$-5 > 85 \times$$

-5	76	31	13	45	26	85	103	99
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$\overset{i}{\underset{\cdot}{\nearrow}}$ $\overset{i}{\underset{\cdot}{\nearrow}}$ $\overset{j}{\underset{\cdot}{\nearrow}}$ $\overset{j}{\underset{\cdot}{\nearrow}}$ $\overset{j}{\underset{\cdot}{\nearrow}}$ $\overset{j}{\underset{\cdot}{\nearrow}}$ $\overset{j}{\underset{\cdot}{\nearrow}}$ $\overset{j}{\underset{\cdot}{\nearrow}}$ $\overset{j}{\underset{\cdot}{\nearrow}}$

$a[i] \leq \text{pivot}$

$-5 \leq -5$ ✓

$76 \leq -5$ ✗

$a[j] > \text{pivot}$

$99 > -5$ ✓

$103 > -5$ ✓

$85 > -5$ ✓

$26 > -5$ ✓

$45 > -5$ ✓

$13 > -5$ ✓

$31 > -5$ ✓

$76 > -5$ ✓

-5	76	31	13	45	26	85	99	103
----	----	----	----	----	----	----	----	-----

$\overset{i}{\underset{\cdot}{\nearrow}}$ $\overset{i}{\underset{\cdot}{\nearrow}}$ $\overset{i}{\underset{\cdot}{\nearrow}}$ $\overset{i}{\underset{\cdot}{\nearrow}}$ $\overset{i}{\underset{\cdot}{\nearrow}}$ $\overset{j}{\underset{\cdot}{\nearrow}}$ $\overset{j}{\underset{\cdot}{\nearrow}}$ $\overset{j}{\underset{\cdot}{\nearrow}}$

$a[i] \leq \text{pivot}$

$76 \leq 76$ ✓

$31 \leq 76$ ✓

$13 \leq 76$ ✓

$45 \leq 76$ ✓

$26 \leq 76$ ✓

$a[j] > \text{pivot}$

$26 > 76$ ✗

$a[i] \leq \text{pivot}$

$99 \leq 99$ ✓

$103 \leq 99$ ✗

-5	26	31	13	45	76	85	99	103
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$\overset{i}{\underset{\cdot}{\nearrow}}$ $\overset{i}{\underset{\cdot}{\nearrow}}$ $\overset{j}{\underset{\cdot}{\nearrow}}$ $\overset{j}{\underset{\cdot}{\nearrow}}$

$a[i] \leq \text{pivot}$

$26 \leq 26$ ✓

$31 \leq 26$ ✗

$a[j] > \text{pivot}$

$45 > 26$ ✓

$13 > 26$ ✗

-5	26	13	31	45	76	85	99	103
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$\overset{i}{\underset{\cdot}{\nearrow}}$ $\overset{i}{\underset{\cdot}{\nearrow}}$

$$a[i] \leq \text{pivot} \quad a[j] > \text{pivot}$$

$$26 \leq 26$$

$$13 > 26 \times$$

$$13 \leq 26 \times$$

~~-5~~ ~~26~~ 13

-5	13	26	31	45	76	85	99	103
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⑤

350	790	400	200	100	96	900	890	760
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$$a[i] \leq \text{pivot}$$

$$350 \leq 350$$

$$790 \leq 350 \times$$

$$a[j] > \text{pivot}$$

$$760 > 350$$

$$890 > 350$$

$$900 > 350$$

$$96 > 350 \times$$

350	96	400	200	100	790	900	890	760
-----	----	-----	-----	-----	-----	-----	-----	-----

$$a[i] \leq \text{pivot}$$

$$350 \leq 350$$

$$96 \leq 350$$

$$400 \leq 350 \times$$

$$a[j] > \text{pivot}$$

$$760 > 350$$

$$890 > 350$$

$$900 > 350$$

$$790 > 350$$

$$100 > 350$$

350	96	100	200	400	790	900	890	760
-----	----	-----	-----	-----	-----	-----	-----	-----

$$a[i] \leq \text{pivot}$$

$$a[j] > \text{pivot}$$

200	96	100	350	400	790	900	890	760
i	j	i		j	i	j		i

$a[i] < \text{pivot}$ $a[j] > \text{pivot}$ $400 < = 400$ $760 > 400$
 $200 < = 200$ $100 > 200$ $790 < = 400$ $890 > 400$
 $96 < = 200$ $\times$ $\times$ $900 > 400$
 $100 < = 200$ $\times$ $760 > 400$

100	96	200	350	400	790	900	890	760
i	j				i	j		i

$100 < = 100$ $96 > 100$ $790 < = 790$ $760 > 790$
 $96 < = 100$ $\times$ $900 < = 790$ $890 > 790$
 $\times$ $900 > 790$

96	100	200	350	400	760	900	890	790
						i	j	i

$900 < = 900$ $790 > 900$
 $890 < = 900$ $790 > 900$
 $790 < = 900$

96	100	200	350	400	760	790	890	900
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